

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

8th Semester of B. Tech. Examination (CL)

May 2014

CL 405 Estimation, Costing & Financial Concepts -II

Date: 05.05.2014, Monday

Time: 01:30 p.m. To 4:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q-1 Figure 1 & Table-1. Shows the details of a steel stanchion with grillage foundation [07]
prepare detail estimate
- Q-2(A) Workout the quantity of foundation work up to plinth level, for drawings given in [14]
Figure 5a & 5b. Refer Table-2. Prepare Abstract for same.

OR

- Q-2(A) Workout the quantity of following items for Ground floor, drawings given in Figure [14]
5a. (i) R.C.C work (ii) Flooring work (iii) Form work Refer Table-1. Also calculate
quantity of steel, cement sand and Aggregate for R.C.C work in super structure.
- Q-3(A) What is project? How it is different from non-project. [05]
- Q-3(B) Workout quantity of septic tank and soak pit shown in Figure 2 [09]

OR

- Q-3(A) Give general guide lines for report preparation of any civil work. Also include the [05]
process of documentation of report.
- Q-3(B) Workout the quantity of sanitation and water supply works, from Figure 6. [09]

SECTION - II

- Q-4 (A) Work out quantities of excavation, P.C.C, R.C.C and brick work, for village road [07]
culvert shown in Figure 3.
- Q-5 Attempt any Two [14]
- Q-5(A) Explain following difference forms of value
i) Book value ii) Distress Value iii) Market Value iv) Monopoly Value v) Replacement
value vi) Salvage value vii) Scrap value
- Q-5(B) Calculate the depreciated replacement cost of building having following particulars by
adopting straight line method and sinking fund method.
Total built-up area of all the floors.....350 m²
Age of building.....25 years
Total life of building.....70 years
Scrap value at the end of useful life.....10%
Percentage of sinking fund5%
Assume present rate of construction at Rs. 1300/- per m².
- Q-5(C) Differentiate Mortgage and lease.
- Q-6(A) Distinguish between Earnest money and Security money. [04]

- Q-6(B) Estimate the cost of earthwork for a portion of road for chainage 10 to 20. Length of chain is 40m. The formation level at 10th chainage is 52m and road is downward gradient of 1 in 200. Formation width of the road is 10m. Side slopes are 2:1 in banking 1.5:1 in cutting. [10]

Chainage	10	11	12	13	14	15	16	17	18	19	20
R.L. of ground	51	50.9	50.5	50.8	50.6	50.7	51.2	51.4	51.3	51	50.6

OR

- Q-6(A) Write general condition of contract. [04]

- Q-6(B) Prepare detail estimate for the construction of a macadam road for one kilometers having following specifications and cross section shown in Figure 4. Formation width of road – 10 m, Average height of bank – 80 cm. Side slope of bank – 2:1(H:V). Metalled width of road – 3.7 m. Soiling shall be over burnt bricks laid flat. Inter coat of metalling shall be of stone ballast 12 cm. thick loose layer compacted to 8 cm. Top coat of metalling shall be of stone ballast of 9 cm. thick loose layer compacted to 6 cm. Side shall be provided with brick on end edging. For bridge and culvert, and other miscellaneous items, provide lump sum amount of 35,000/- per kilometer. Assume rates prevailing to your locality. [10]

Table 1.0 Weight of steel section(Q.1)

Section Name	Size	Wt/unit
Upper tier-ISLB 500	500mm x 180mm	@ 75 kg per meter
Lower tier-ISLB 400	400mm x 165mm	@59.90kg per meter
Stanchion-ISLB 450	450mm x 250mm	@177.89kg per meter
Side angle and web angle ISA 150x 150 x 10	150mm x 150mm x 10mm	@22.80kg per meter
Base plate	1.0m x 1.0m x 12mm	Calculate wt per sqm
Gusset Plate	As per drawing 12mm thick	by assuming wt of steel per cum as 7850kg

The steel work is embedded in concrete of 1:1.5:3

Table 2.0 Details of columns & Footings (Q-2)

Column Group	A1,A2,A3,A4	B1,C1,B4,C4 ,D1,E1,D4,E4	F1,F2,F3,F4	Column Height 3.5 m
P.C.C	1010 X 2600	5490 X 1700	1200 X 1800	
Footing	910 X 2400	5290 X 1500	1000 X 1600	
Column base	390 X 760	800 X 430	390 X 760	
Column	230 X 600	600 X 230	230 X 600	
Plinth Beam	230 X 300	External Beam	230 X 450	
Internal Beam	230 X 700	Slab Thickness	125mm	

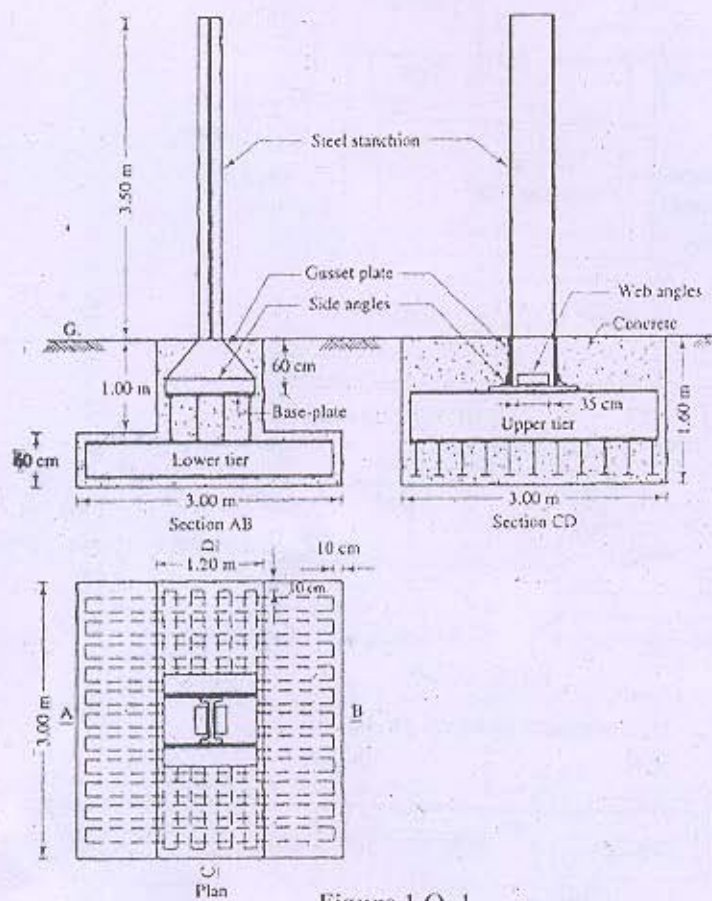


Figure 1 Q.1

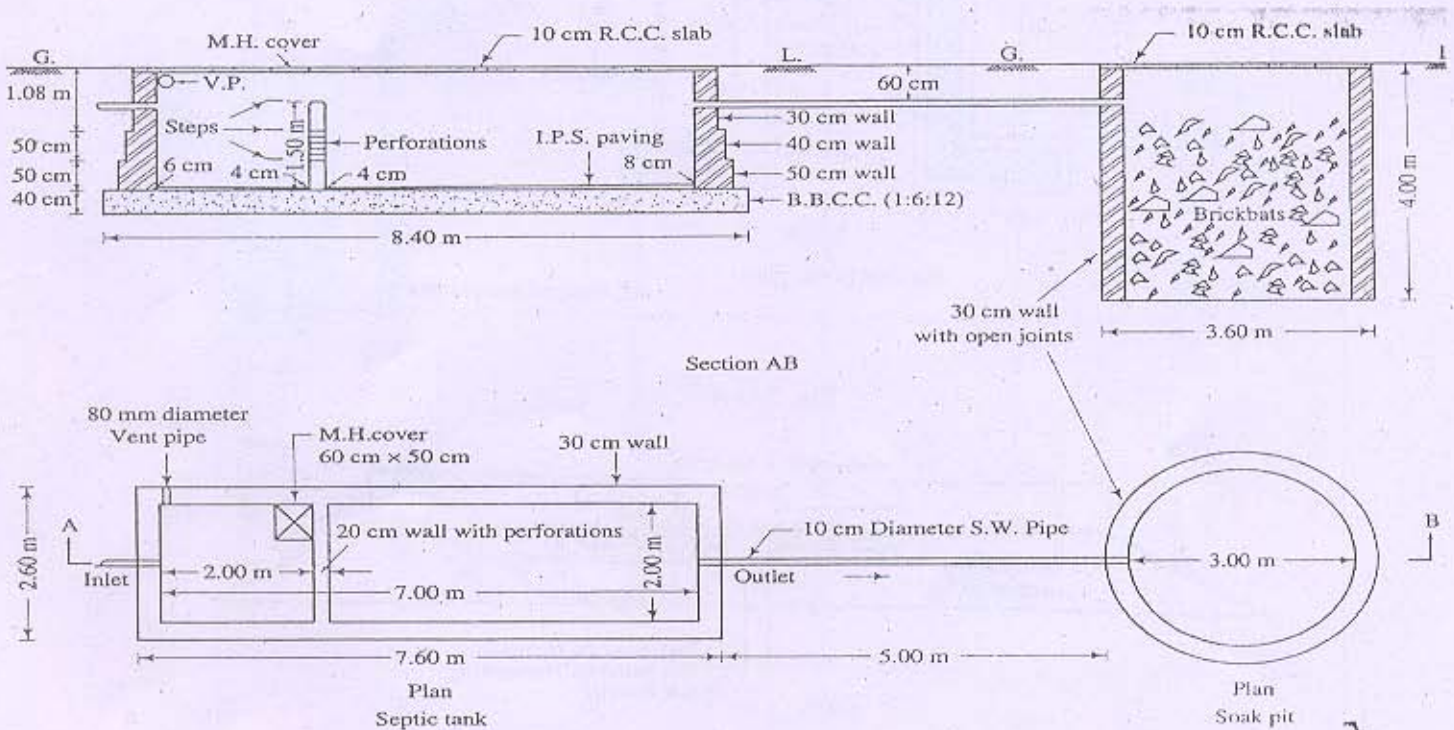


Fig. 2 Q-3 (B)

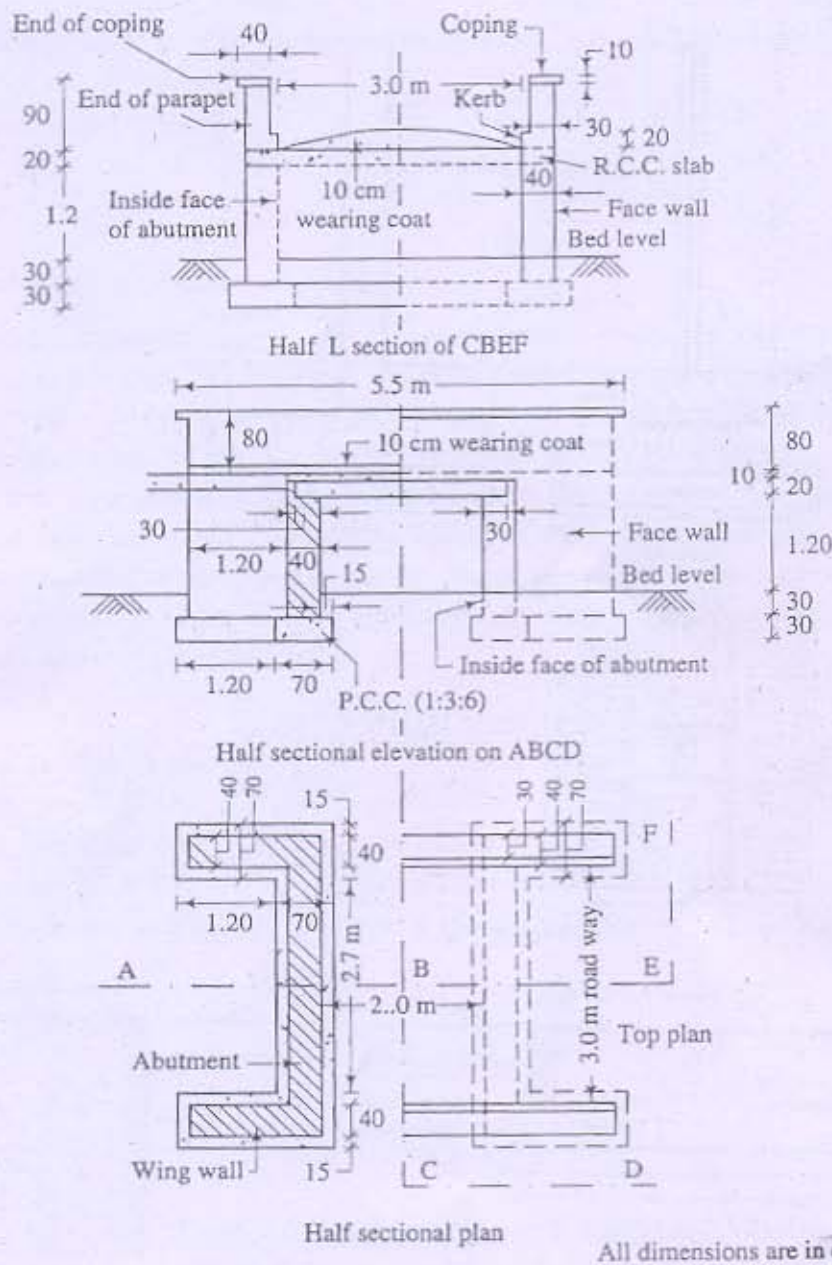


Fig. 3 Q-4

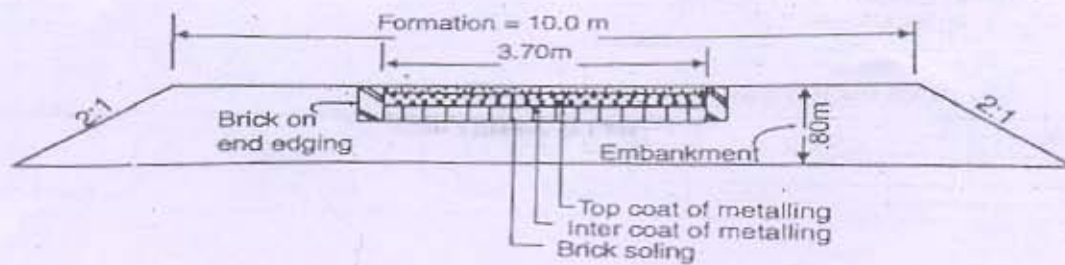
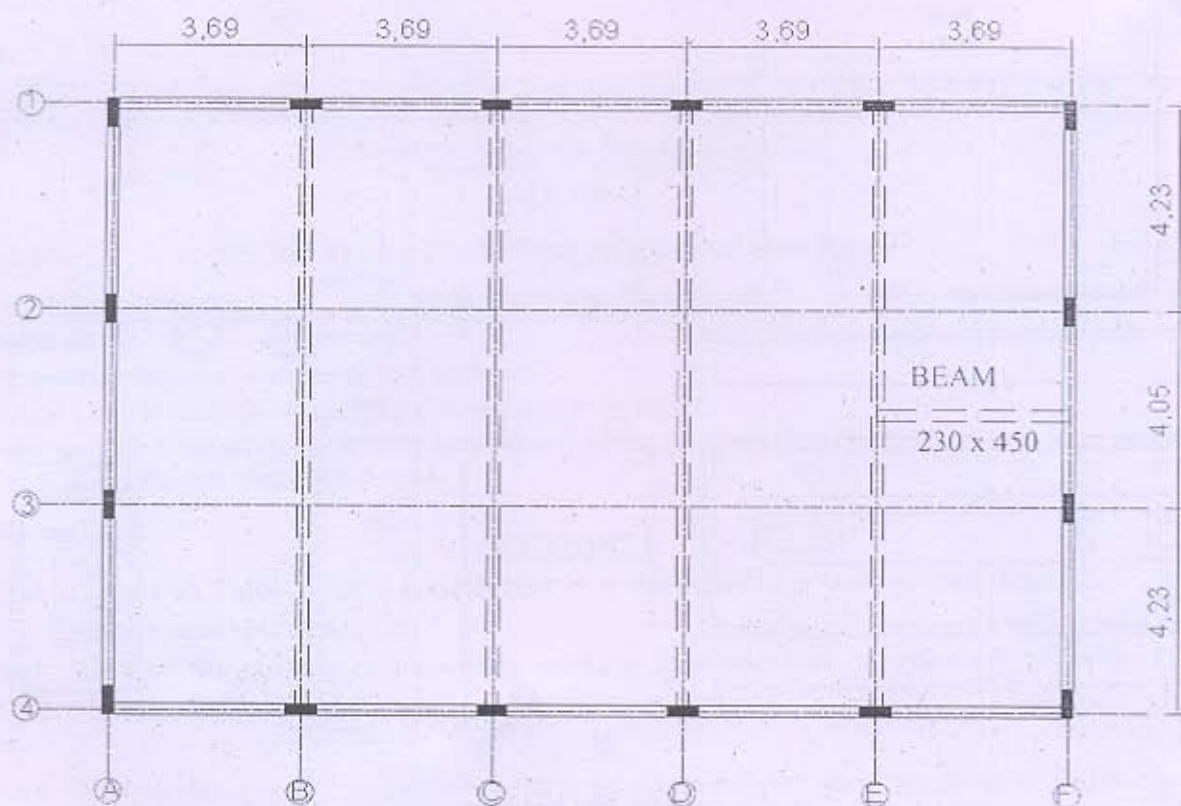
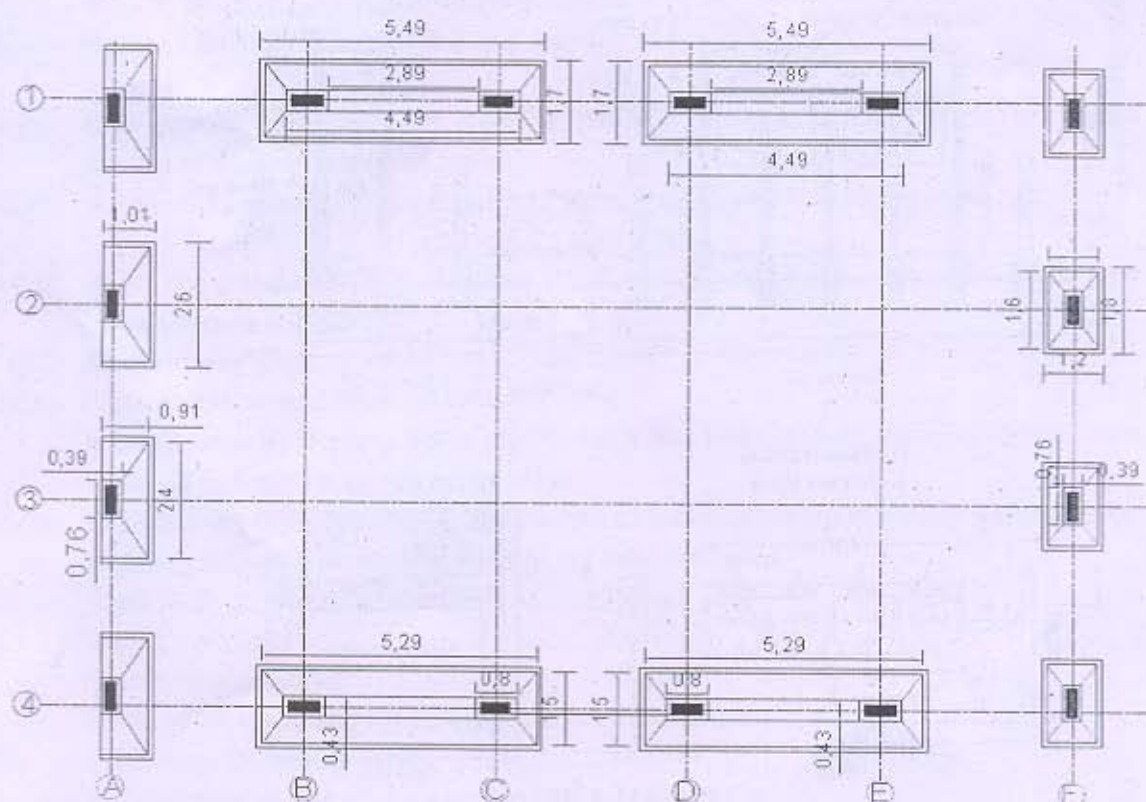


Fig. 4 Q-6(B)



STRUCTURAL LAYOUT OF COMMUNITY HALL



FOUNDATION LAYOUT

Fig. 5a Q-2(A)

